Defining a paleomagnetic polarity pattern in the Monteregian intrusives

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Oka and nearby small plutons on the western end of the Monteregian Hills were sampled for paleomagnetic study at 43 sites (569 specimens). Every specimen was AF step demagnetized in 4 kA/m increments to 20 or 24 kA/m. Consistent remanence directions were found for 36 sites (452 specimens). Use of a stability index to select only those specimens with the best defined end points does not improve the site statistics. The Oka, Brilund, Carillon, and Ile Cadieux plutons have statistically similar mean remanence direction populations which are different from the Ste. Dorothée sill direction. Except for one Carillon site, all site mean directions are normally polarized, whereas all nine plutons, except for Mt. Johnson, from the middle and eastern end of the Monteregian Hills are reversely polarized. Normally and reversely polarized plutons give statistically similar but antiparallel pole positions, giving a combined pole position of 169.0°W, 72.4°N ($\delta_p = 2.8°, \delta_m = 3.7°$), which is consistent with the 120 ± 4 Ma radiometric age. The polarity pattern evidence suggests that Oka and adjacent plutons were emplaced rapidly during one normal polarity interval, and that the Monteregian Hills plutons were emplaced progressively from west to east during two normal and two reversed polarity intervals lasting ~ 2 Ma. This leads to some speculations on the plume and rift modes of emplacement.

On a échantillonné le pluton d'Oka et d'autres plutons voisins plus petits dans la partie ouest des collines montérégiennes pour une étude paléomagnétique à 43 sites (569 spécimens). On a désaimanté chaque échantillon dans un champ alternatif par étapes avec des paliers de 4 kA/m jusqu'à 20 ou 24 kA/m. On a trouvé des valeurs de rémanence compatibles pour 36 sites (452 spécimens). L'utilisation d'un indice de stabilité pour choisir seulement ces spécimens avec les valeurs terminales les mieux définies n'améliore pas les statistiques d'un site donné. Les plutons d'Oka, de Brilund, de Carillon et d'Ile Cadieux ont des populations de directions moyennes de rémanence statistiquement semblables qui diffèrent de la direction obtenue pour le filon-couche de Sainte-Dorothée. A l'exception d'un site dans le pluton de Carillon, toutes les directions moyennes d'un site sont normalement polarisées alors que les neuf plutons—à l'exception du mont Johnson—de la partie moyenne ou est des Montérégiennes ont des polarités inversées. Les plutons à polarité normale ou inverse définissent des directions de pôles statistiquement semblables mais antiparallèles, ce qui donne une position combinée pour le pôle à 169.0°O, 72.4°N ($\delta_p = 2.8°, \delta_m = 3.7°$), position qui est compatible avec l'âge radiométrique de 120 ± 4 Ma. Les preuves fournies par les types de polarité suggèrent que le pluton d'Oka et ses voisins ont été mis en place rapidement durant un intervalle de polarité normale et que les plutons des Montérégiennes ont été mis en place progressivement d'ouest en est durant deux intervalles de polarité normale et deux de polarité inverse sur une période d'environ 2 Ma. Ce modèle conduit à certaines spéculations sur les modes de mise en place par panache chaud et fissure.

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Introduction

The Monteregian Hills form a petrologic province or belt of Cretaceous basic alkalic intrusions cutting Paleozoic and Precambrian rocks in southwestern Quebec. Larochelle (1961, 1968*a*, 1969) has reported paleomagnetic data from 10 plutons. No data have been reported for plutons on the western end of the belt except for his three sites from Oka (Larochelle 1969). Curiously, Oka retained a normal polarity along with Mt. Johnson, whereas the other eight plutons gave reversed remanence directions. This study of Oka and the small nearby Ste. Dorothée, Brilund, Ile Cadieux, and Carillon plutons was undertaken to examine these plutons on the western end of the belt in terms of their contemporaneity and of their relationship to the other Monteregian Hills. A secondary objective was to see if progressive alternating field (AF) demagnetization in 4 kA/m increments of *all*

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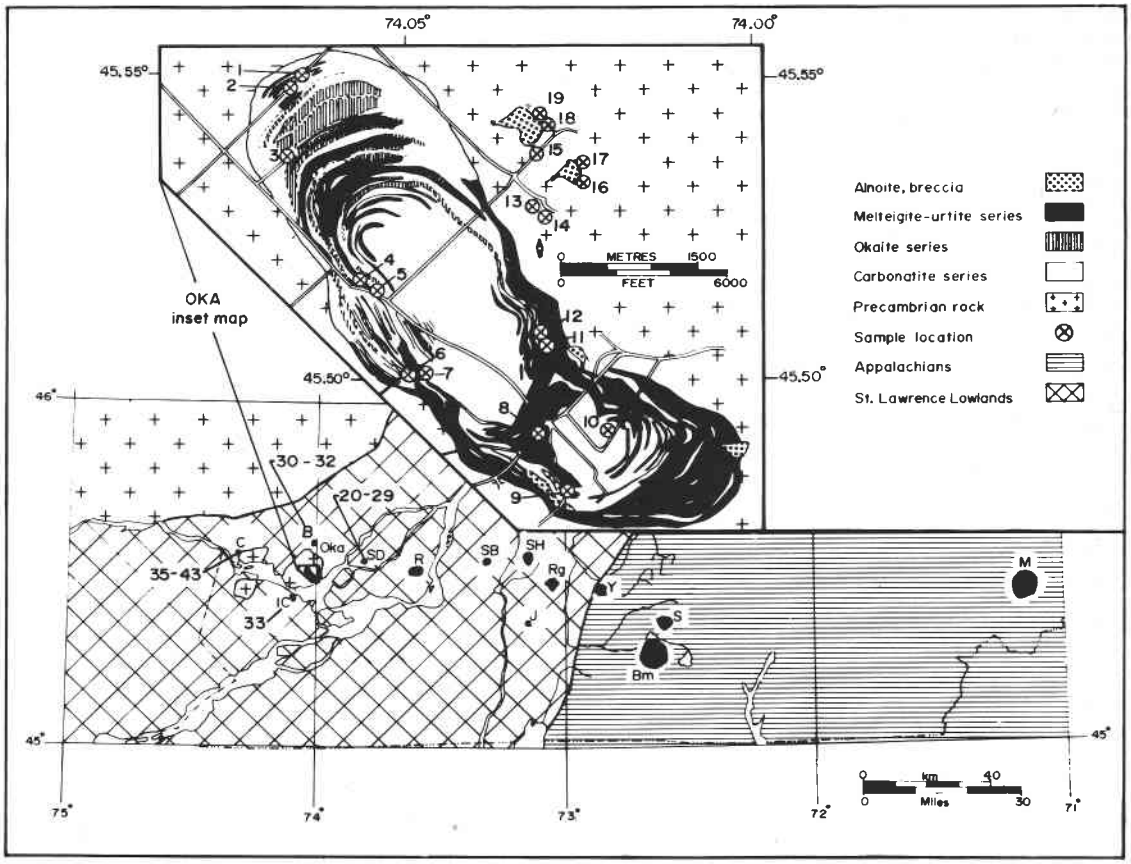


FIG. 1. Sample locations and geology of the Monteregian Hills, with an inset map of Oka from Gold (1966). The pluton abbreviations are: C = Carillon, IC = Ile Cadieux, Oka, B = Brilund, SD = Ste. Dorothée, R = Mt. Royal, SB = Mt. St. Bruno, SH = Mt. St. Hilaire, J = Mt. Johnson, Rg = Mt. Rougemont, Y = Mt. Yamaska, Bm = Mt. Brome, S = Mt. Shefford, and M = Mt. Megantic.

specimens would significantly improve the definition of the pluton mean remanence vectors.

Geology

Monteregian Hills

The geology of the Monteregian Hills has been extensively discussed (Adams 1903; Graham 1944; Pouliot 1962; Gold 1967; Gold and Marchand 1969). These Cretaceous basic alkalic plutons are distributed along an east–west trending belt that is ~250 km long by 50 km wide (Fig. 1). It coincides with the eastward extension of the Ottawa graben across the St. Lawrence Lowlands and into the Appalachians, and it may be part of a 1500 km zone of intermittent alkalic intrusion over the past 600 Ma (Kumarapeli and Saull 1966). The fault pattern within the belt appears to control the location of the plutons (Philpotts 1969). The host rocks include Grenville granitic gneisses in inliers at the west end of the belt, flat-lying Cambro-Ordovician sediments in the Lowlands, and folded lower Paleozoic sediments in the Appalachians at the east end.

Going from west to east along the belt, the Monteregian plutons display a number of trends. Their bulk composition begins as ultrabasic–basic in the west end and becomes progressively more silicic to the east (Gold 1967). Their mafic content decreases from >90% in Mt. St. Bruno and westward to <40% in Mt. Shefford and eastward. The mafic rock types progress from alnoites, essexites, peridotites, and pyroxenites bearing titanaugite, to gabbro and diorite bearing hornblende and augite. The associated leucocratic rocks progress from ijolites and carbonatites bearing melilite, to low quartz syenites bearing analcite, and to normal quartz differentiates including granites and bearing Ca plagioclase. Similarly, the plutons change form. Explosive intrusion forming alnoite diatreme breccia pipes, small plugs, and dikes

with brecciated country rock occur in the west (Philpotts 1969). Passive intrusion becomes increasingly important and dominates in the east, giving differentiated zoned plutons surrounded by dikes, ring dikes, cone sheets, and sills.

The Monteregian plutons give K–Ar and Rb–Sr radiometric age dates in the 112 ± 12 Ma range with the more reliable isochrons in the 120 ± 4 Ma range (Fairbairn *et al.* 1963; Gold 1967; Shafiqullah *et al.* 1969; using decay constants for K–Ar of 5.30×10^{10} year^{-1} and for Rb–Sr of 1.39×10^{-11} year^{-1}).

Oka

The Oka complex is 8 km long by 2 km wide and shaped like an "8" (Fig. 1). It intrudes Precambrian gneisses which are fenetized by contact metamorphism (Rowe 1958; Gold 1966, 1969). The lobes of the "8" are defined by concentric lithostructural ring dikes or cone sheets of melteigite–urtite series rocks which dip steeply outwards except in the centre of the northern lobe. Large concordant lunes of okaite series rocks are found around the northern end of the complex. Irregular zones and dikes of alnoite and alnoite breccia form concordant and discordant dikes, plugs, and pipes, both in the complex and in the adjacent host rocks. The carbonatite lobe-centres are comprised of flow-foliated calcite with thin ferromagnesian-magnetite bands and occasional biotite–serpentine lenses. This rock grades into apatite-bearing dolomite around the northern end. Thin late lamprophyre dikes cut both the complex and host rocks. Shafiqullah *et al.* (1969) obtained a K–Ar isochron on 12 micas of 120 ± 4 Ma which agrees with the average of three mica Rb–Sr ages of 121 ± 7 Ma obtained by Fairbairn *et al.* (1963). Pyrochlore is mined from the complex near LaTrappe.

Ste. Dorothée

Seven fourchite sills up to 7 m thick intrude the Beekmantown dolomites near Ste. Dorothée (Clark 1952, 1972; Gold and Marchand 1969). Nearby are three diatreme breccia pipes. The sills are differentiated and grade upward from the basal aphanitic chilled margin through zones of vertical camptonitic stringer ocelli, hornblende–augite phenocrysts, augite phenocrysts, hornblende phenocrysts, vesicules or amygdules, horizontal stringer ocelli, and finally the uppermost chilled margin.

Brilund

The vertical Brilund ultrabasic alnoite pipe is ~175 m in diameter and cuts Potsdam sandstone (Gold and Marchand 1969). It is comprised of coarse-grained, blue–gray poikilitic monticellite with some brecciation around the margin and some alteration in veins.

Ile Cadieux

The Ile Cadieux alnoite pipe is similar to the Brilund pipe except that it cuts Beekmantown dolomites and has a slightly larger diameter (Bowen 1922).

Carillon

The Carillon or St. André carbonatite is a small plug ~1000 m long by ~500 m wide cutting Precambrian gneisses.

Procedure

At each of the 43 sites, a series of cores were drilled and oriented *in situ* to within ~2° using a solar compass. Except for a few single specimen cores, two right-cylindrical 2.50 cm specimens were cut from each core yielding a total of 569 specimens (Table 1). The natural remanent magnetization (NRM) of each specimen was measured on a Schonstedt SSM-1A spinner magnetometer. *Every* specimen was AF demagnetized in steps at 4, 8, 12, 16, 20, and, for about half the specimens, 24 kA/m (Table 1) using a Schonstedt GSD-1 demagnetizer and its remanence remeasured after each step.

AF Step Demagnetization Results

Virtually all of the specimens behaved in the same manner on AF demagnetization. Figure 2 shows the decay of the site mean intensities for a representative one-third of the sites. Most specimens have NRM intensities ranging from ~0.1–20 A/m. They undergo rapid, roughly exponential decay up to ~12 or 16 kA/m as fairly stable, normal, viscous remanent magnetizations (VRM) are removed. Most curves then flatten to indicate a slower exponential decay rate as a more stable, normal, remanent magnetization is uncovered. At 20 or 24 kA/m this stable remanence is generally only 5–10% of the NRM intensity. The stable remanence is believed to be primary and acquired during cooling of the intrusions as a thermoremanent magnetization (TRM). The remanence may also be, in part, a primary chemical remanent magnetization (CRM), partial TRM or partial CRM.

The NRM directions generally cluster around the Recent direction of 0°, 65°. Upon demagnetization they swing anticlockwise to their primary direction. A few examples are shown in Fig. 3. Reversed remanence directions were found in one or two specimens from each of four sites (17, 18, 32, and 37) and all specimens from one site (36). These

TABLE 1. Site mean remanence directions after AF demagnetization

Site	H	Directions of all specimens					Directions of specimens with PSI < 100 μdeg m/A				
		N	Decl.	Incl.	α_{95}	k	N	Decl.	Incl.	α_{95}	k
1	20	25	338.9	51.0	6.8	19.0	9	340.4	61.5	6.6	62.0
2	20	18	337.6	65.1	2.6	177.6	14	337.9	64.7	2.5	254.6
3	20	12	353.1	67.6	9.7	20.9	3	357.8	61.7	20.6	36.7
4	24	12	331.6	51.9	9.3	22.8	12	331.6	51.9	9.3	22.8
5	24	14	343.3	52.6	8.0	25.8	14	343.3	52.6	8.0	25.8
6	20	11	325.1	66.6	2.6	319.4	11	325.1	66.7	2.6	319.4
7	10	10	321.1	62.6	4.2	132.3	10	321.1	62.6	4.2	132.3
8	24	6	345.1	55.7	2.8	565.0	6	345.1	55.7	2.8	564.9
9	20	14	14.9	13.3	4.7	72.3	14	14.9	13.3	4.7	72.3
10	24	12	314.6	71.0	27.9	3.4	0				
11	24	30	55.0	77.6	17.2	3.3	6	119.4	52.7	48.5	2.9
12	24	15	24.0	64.5	15.2	7.3	3	332.6	57.5	41.5	9.9
13	20	12	339.9	58.6	5.7	59.3	12	339.9	58.6	5.7	59.3
14	20	14	343.8	58.6	3.9	103.4	14	343.8	58.6	3.9	103.4
15	20	14	347.4	46.9	9.7	17.9	14	347.4	46.9	9.7	17.9
16	24	14	327.7	70.3	7.9	26.6	12	328.4	69.2	9.0	24.2
17	24	14,2	350.8	68.4	2.7	190.1	14,2	350.8	68.4	2.7	190.1
18	20	13,1	343.6	71.3	18.0	5.9	13,1	343.6	71.3	18.0	5.9
19	20	14	340.2	24.2	22.6	4.1	14	340.2	24.2	22.6	4.1
20	24	11	327.6	56.5	2.4	368.3	9	327.8	56.6	2.8	340.3
21	24	11	323.6	59.0	3.1	222.3	0				
22	24	15	317.2	53.2	4.2	83.8	0				
23	24	12	318.0	53.7	2.7	270.0	5	314.5	53.7	3.4	514.3
24	24	8	324.6	51.0	5.4	107.2	0				
25	24	11	335.9	56.5	9.6	23.6	9	336.9	51.4	5.3	97.3
26	24	7	332.5	56.4	9.3	42.9	5	326.3	58.1	8.5	83.0
27	24	6	336.5	59.9	4.7	207.0	0				
28	24	10	329.2	49.8	4.7	107.0	5	331.9	50.3	8.6	81.0
29	24	9	335.4	54.8	3.4	229.6	3	328.7	56.1	12.2	103.0
30	24	9	329.1	70.0	9.2	32.1	9	329.0	70.0	9.2	32.1
31	24	10	327.5	67.0	6.7	53.1	8	317.8	65.8	5.2	113.1
32	24	16,2	310.3	69.3	20.8	3.7	11,1	322.8	71.2	20.7	5.4
33	24	12	356.8	74.0	11.0	16.5	2	351.3	69.2	51.6	6.4
34	20	16	341.1	73.4	2.2	281.5	12	343.0	72.8	2.7	251.9
35	20	15	343.4	62.8	9.9	16.1	3	352.5	66.2	42.3	9.5
36	20	18	160.8	−36.8	4.3	65.1	9	160.0	−38.8	3.7	199.9
37	20	10,2	347.0	60.0	13.1	12.0	6,2	344.7	51.8	6.1	82.3
38	20	10	332.5	65.7	14.3	12.5	5	329.2	61.6	25.7	9.8
39	20	16	143.1	80.8	4.6	66.8	5	131.7	83.2	10.1	58.7
40	20	16	143.4	83.9	5.3	50.1	5	51.8	83.8	8.0	93.5
41	20	14	347.1	75.8	3.6	124.3	11	344.3	76.1	4.4	107.6
42	20	10	328.9	69.1	2.9	275.1	10	328.9	69.1	2.9	275.1
43	20	16	19.0	39.5	5.0	54.6	0				
		569 = 100%					333 = 59%				

NOTES: H is the AF demagnetizing field in kA/m. N is the number of specimens with normal directions and with reversed directions following the comma. Declination (Decl.), inclination (Incl.), and the radius of the cone of 95% confidence (α_{95}) are in degrees with + down. k is Fisher's (1953) precision parameter. The locations of the sites are as follows: Oka (sites 1–19; 74.04°W, 45.51°N), Ste. Dorothée (sites 20–29; 73.83°W, 45.50°N), Brilund (sites 30–32; 74.01°W, 45.61°N), Ile Cadieux (site 33; 74.05°W, 45.25°N), and Carillon (sites 34–43; 74.27°N, 45.57°W).

showed the expected behaviour in direction and intensity for the removal of normal VRM's from antiparallel reversed TRM's. The movement of the remanence direction of each specimen was analysed by calculating the paleomagnetic stability index (PSI) (Symons and Stupavsky 1974). For a majority of specimens the PSI values are high (i.e., >2500 μdeg m/A) at 0, 4, and 8 kA/m, and then progressively decrease to relatively low values of ~600 ± 300 μdeg m/A at 20 or 24 kA/m. A total of 333 specimens, or 59%, give PSI values of <1260 μdeg m/A (i.e., <100 m deg/Oe) at their maximum cleaning field.[2]

[2]SI units: 1 Oe = 0.0796 kA/m; 1 emu/cm³ = 10^3 A/m; 1 m deg/Oe = 12.6 μdeg m/A.

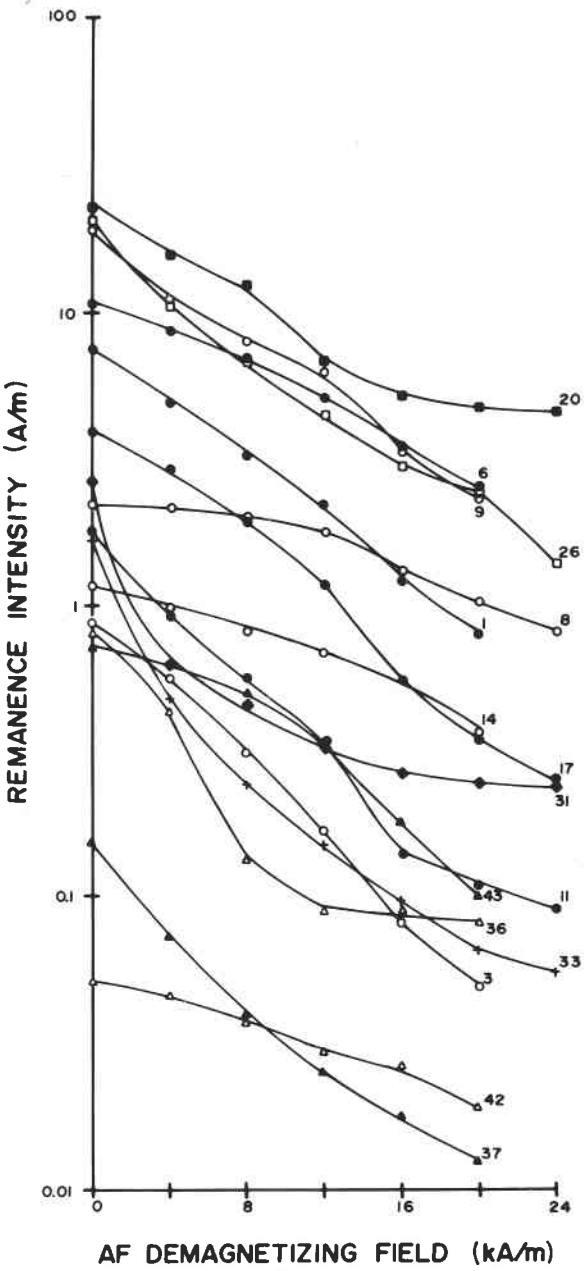


FIG. 2. AF step demagnetization curves of site mean remanence intensity in amperes/metre against field in kiloamperes/metre.

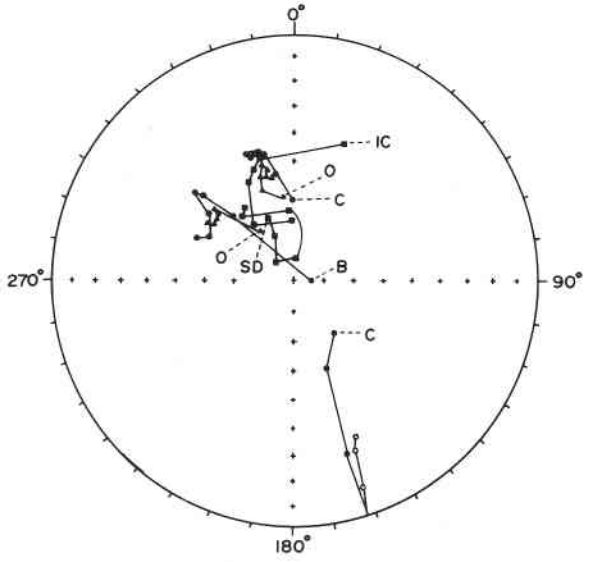


FIG. 3. AF step demagnetization curves for typical specimens' remanence directions plotted on equal-area stereonet with solid symbols indicating normal (down) vectors and open symbols indicating reversed (up) vectors.

Site Mean Directions

The reversed remanence directions of specimens in dominantly normal sites were changed to their antiparallel position. The site mean remanence directions were then computed for the maximum 20 or 24 kA/m AF demagnetizing treatment, giving all specimen vectors unit weight (Table 1). The site mean remanence directions were then recomputed using only specimen vectors with PSI values of < 1260 μdeg m/A. Theoretically, this screen should let pass only those specimens with the best-defined stable "end-point" directions.

Pluton Mean Directions

Only sites having a radius of 95% confidence (α_{95}) (Fisher 1953) of less than 15° were deemed to be sufficiently well defined to be considered for further analysis (Fig. 4). The polarity of the site mean remanence direction was reversed for site 36. The accepted site mean directions were then given unit weight and combined to give the intrusive mean remanence direction both before and after applying the PSI screen (Table 2). Site 9 from the Oka instrusion has an intermediate remanence direction which lies well outside the cone of three times the circular standard deviation about the Oka mean. Therefore, this site was excluded and the Oka intrusive mean was then recalculated.

The variance ratio test was then used to compare the "unscreened" and "screened" intrusive means (Table 2) (Larochelle 1968*b*). The screening process does not significantly improve the clustering of any of the four intrusives. For two intrusives, Oka and Carillon, there is a slight improvement, and for two intrusives, Ste. Dorothée and Brilund, there is a slight deterioration in the cluster. Clearly the improvement of the cluster statistics accomplished by removing the deviantly directed specimens with the PSI screen is about offset by the deterioration

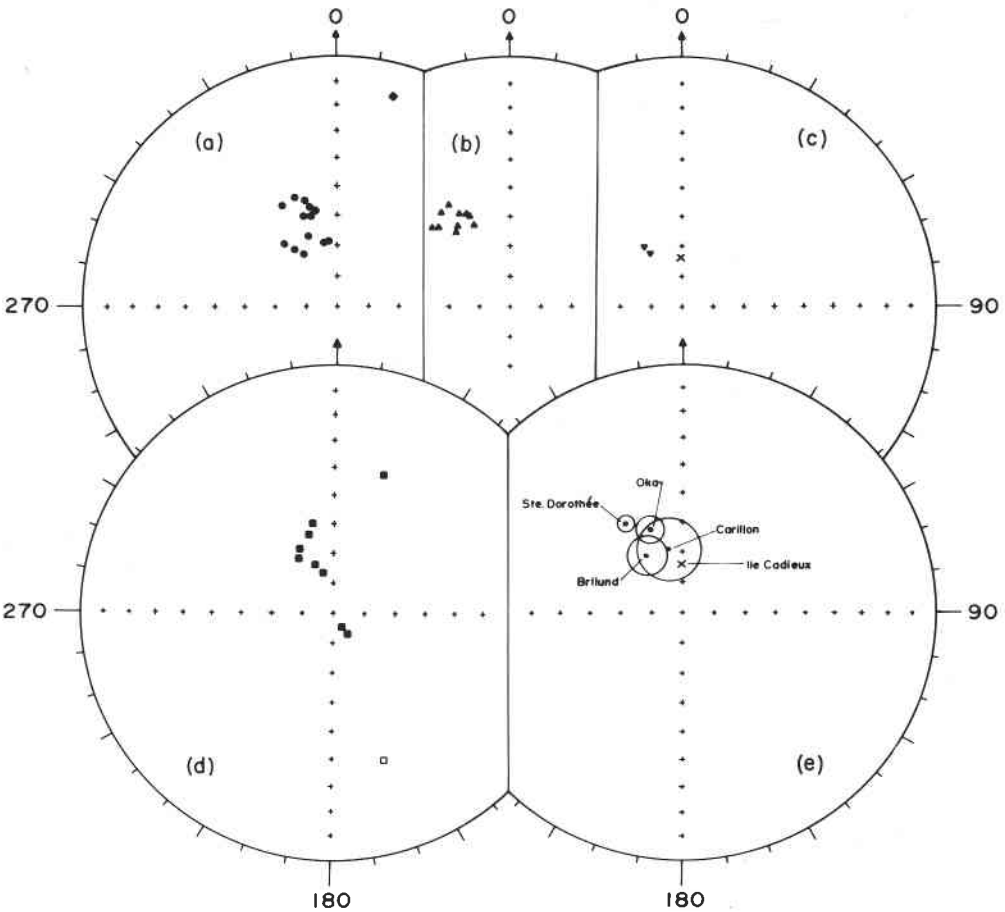


FIG. 4. Site mean remanence directions for the (*a*) Oka, (*b*) Ste. Dorothée, (*c*) Brilund and Ile Cadieux (X), and (*d*) Carillon pluton sites. The pluton mean directions are shown in (*e*) along with the cone of 95% confidence except for the single site of Ile Cadieux. Plotting conventions as in Fig. 3.

caused by removing a large number of the concordantly directed specimens with high PSI values, i.e., the improvement in the individual specimen quality in the population is offset by the decrease in the specimen population. Accordingly, the unscreened data are used, hereafter, in this paper.

Comparison Between Plutons

Next, a series of variance tests were run to compare the site mean remanence direction populations of each of the plutons, excluding Ile Cadieux which has only one reliable site (Table 3). For each pair, the ratio of the variances in dispersion ($V = \delta_A^2/\delta_s^2$) or ($V = \delta_s^2/\delta_A^2$) was calculated and compared to the corresponding theoretical statistic $F_{2(B-1),\,2(N-B),\,0.05}$. If $V > F$, then the two populations have significantly different distributions and hence probably reflect different conditions of remanence acquisition or removal. The second test is more powerful. It computes the ratio of the angular variances between the two populations to those within the two populations ($F_c = \delta_b^2/\delta_w^2$) and compares this to the theoretical statistic $F_{2,(B-2),\,0.05}$. If $F_c > F$, then the two populations define significantly different directions and hence are not contemporaneous. The results of these tests are fairly straight forward. The Ste. Dorothée sill is not contemporaneous or related in mode of remanence acquisition to any of the other three plutons. This is probably because the sill was intruded as a separate event in terms of time, and (or) because the sill cooled rapidly so that it represents a virtual paleomagnetic pole in which the effects of secular variation have not been sufficiently averaged to prevent bias. Conversely, the Oka, Brilund, and Carillon plutons appear to be contemporaneous and similar in cooling histories. The positive test results for Brilund on the initial test is attributed to the fact that it is represented by only two sites whose directions happen to be very close together. Finally, it appears likely, on the

TABLE 2. Group mean remanence directions and variance tests of screening criteria

Group	B	Decl.	Incl.	α_{95}	k	V	F, 95%	Result
Oka								
A	13	339.3	60.0	4.6	82.0	1.16	2.03	No
S	12	338.6	60.1	4.5	94.8			
Ste. Dorothée								
A	10	327.8	55.3	3.1	243.6	1.07	2.41	No
S	6	327.9	54.6	4.5	226.8			
Brilund								
A	2	328.2	68.5	6.7	1396.2	3.78	19.00	No
S	2	322.9	68.0	13.0	369.4			
Ile Cadieux								
A	1	356.8	74.0	—	—	—	—	—
S	0	—	—	—	—			
Carillon								
A	10	349.6	68.6	12.9	14.9	1.28	2.33	No
S	9	343.2	69.1	11.5	21.2			

NOTES: B is the number of sites. V is the variance ratio δ_A^2/δ_S^2 or δ_S^2/δ_A^2 of two populations. If V is greater than the theoretical parameter F, then the two populations are different at the 95% confidence level. Only sites with $\alpha_{95} < 15°$ are included. Site 36 is reversed to its antiparallel direction. Under "Group", A indicates site means using all specimens and S using only specimens with PSI < 1260 μdeg m/A. Other captions as in Table 1.

TABLE 3. Variance tests between intrusives

Intrusive pair	B	V	F 95%	Result	F_c	F, 95%	Result
Oka – Ste. Dorothée	23	2.97	2.15	Yes	5.82	3.23	Yes
Oka – Brilund	15	17.03	19.45	No	2.22	3.37	No
Oka – Carillon	24	5.00	2.03	Yes	2.65	3.22	No
Ste. Dorothée – Brilund	12	5.73	19.44	No	11.81	3.49	Yes
Ste. Dorothée – Carillon	21	14.86	2.19	Yes	6.64	3.24	Yes
Brilund – Carillon	13	85.17	19.45	Yes	0.29	3.44	No
Normal – Reverse*	70	2.60	1.52	Yes	1.80	3.06	No

NOTES: F_c is the calculated value for $\delta_b^2/\delta_\omega^2$. Other captions and site details are noted in Table 2.
*The normal and reverse statistics are given in Table 4.

basis of its single site, that the Ile Cadieux pluton is contemporaneous with the Carillon pluton.

Comparison to Previous Data

Larochelle (1969) obtained a mean remanence direction of 339.1°, 58.0° ($\alpha_{95} = 4.1°$) for Oka based on data from 24 specimens (12 cores from three sites). His direction is a scant 2.2° and statistically indistinguishable from the 339.3°, 60.0° ($\alpha_{95} = 4.6°$) direction of this study (Fig. 5).

Comparing these results of this study to the data from the rest of the Monteregian Hills is less certain. Only Mt. Johnson yields normal remanence polarity, and its direction of 330.9°, 63.1° is statistically similar to Oka and adjacent plutons on the western side of the Monteregian Hill petrographic province (Fig. 5) where its cone of circular standard deviation (α_{63}, Fisher 1953) intersects all other α_{63} cones.

Next, normally and reversely polarized sites were compared (Table 4). The 41 normal sites include 35 from this study plus six from Larochelle (1968*a*, 1969). The 29 reversed sites include one from this study plus 26 from Larochelle (1968*a*, 1969) plus the data for Mt. Megantic from Larochelle (1961) given a weight of two sites. The variance test between the two populations, after conversion of the reversed sites to normal polarity (Table 3), shows that while they do have significantly different dispersion statistics—a common feature when normal and reverse populations are compared—they do not have significantly

TABLE 4. Mean remanence and pole positions for the Monteregian petrographic province

Group	No. of sites	No. of spec.	Resultant vector (R)	Mean remanence direction				Pole position			
				Decl.	Incl.	k	α_{95}	Long. (°W)	Lat. (°N)	δ_p (deg)	δ_m (deg)
Normal sites	41	504	39.911	337.1	61.8	37.6	3.6	163.9	73.4	4.4	5.6
Reversed sites	29	256	28.729	155.4	−58.1	99.7	2.7	175.1	70.7	3.0	4.1
All sites*	70	760	68.599	336.4	60.3	49.2	2.4	169.0	72.4	2.8	3.7

NOTES: Conventions as in Table 1. For the pole position the longitude and latitude are in degrees west (long.°W) and north (lat.°N), respectively, with δ_p and δ_m being the semi-axes of the oval of 95% confidence along and perpendicular to the site-pole great circle.
*Reversed vectors in antiparallel position.

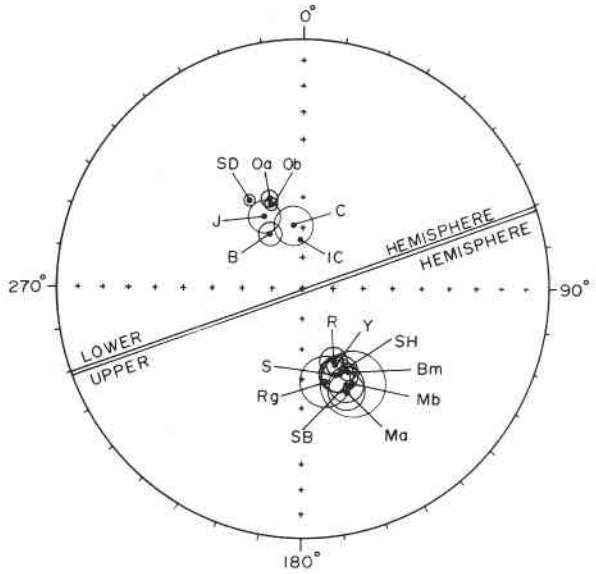


FIG. 5. Pluton mean directions for the Monteregian Hills incorporating data from Larochelle (1961, 1968*a*, 1969). Conventions as in Figs. 1 and 3.

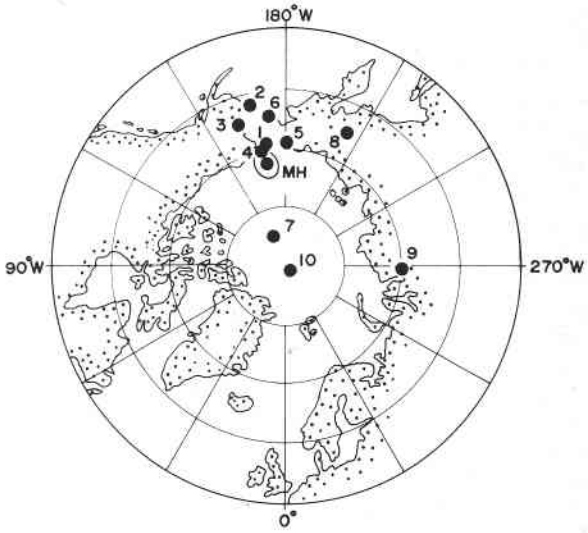


FIG. 6. Pole position for the Monteregian Hills (MH) shown in the northern hemisphere. Additional poles from the North American craton include the: (1) 78 ± 1 Ma Elkhorn Mtn. volcanics (Hanna 1973), (2) 79 ± 10 Ma Niobara limestone (Shive and Frerichs 1974), (3) 80 ± 15 Ma Mesaverde sediments (Kilbourne 1969), (4) 97 ± 10 Ma Arkansas alkalic complex (Scharon and Hsu 1969), (5) 106 ± 5 Ma Isachsen diabase (Larochelle and Black 1963), (6) 130 ± 5 Ma Mt. Ascutney gabbro (Opdyke and Wensink 1966; Foland *et al.* 1971), (7) 136 ± 8 Ma Ithaca peridotite dikes (De Journett and Schmidt 1975), (8) 142 ± 8 Ma Morrison sandstones (Steiner and Helsley 1975), (9) 146 ± 4 Ma Summerville sandstones (Steiner and Helsley 1972), and (10) 158 ± 4 Ma Belknap–Monadnock stocks (Opdyke and Wensink 1966; Foland *et al.* 1971).

different directions. This provides strong evidence indicating that most of the VRM has been removed, that the measured remanences are primary, and that the Cretaceous normal and reversed paleofields were truly antiparallel. Therefore, all the data can be validly combined to form a single pole position.

Pole Position

The combined Monteregian Hills pole position determined from all 70 sites is 169.0°W, 72.4°N ($\delta_p = 2.8°$, $\delta_m = 3.7°$) (Table 4). This pole is about 1.5° away from that reported by Larochelle (1969) and is consistent with an age of 120 Ma based on other pole positions (Fig. 6).

Remanence Polarity

The remanence polarity pattern for the Monteregian Hills is striking. From west to east: the Carillon carbonatite is normal except for one site; the Ile Cadieux, Oka, Brilund, and Ste. Dorothée plutons are normal; Mts. Royal, St. Bruno, and St. Hilaire are reversed; Mt. Johnson is normal; and Mts. Rougemont, Yamaska, Brome, Shefford, and Megantic are reversed. Assuming that the petrologic trends have time significance, it is proposed that intrusion of the Monteregian Hills petrographic province took place first as ultrabasic–basic explosive intrusion at the west end of the belt during a normal geomagnetic interval. The intrusions became progressively more acidic and passive as the centre of activity moved east-

ward with time through a reversed, a normal, and finally a reversed geomagnetic interval. Using Larson and Hilde's (1975) mid-Jurassic to mid-Cretaceous polarity time scale, it is evident that this sort of pattern would not likely be formed after the onset of the mid-Cretaceous "normal" quiet zone starting at ~114 Ma. Thus, the Monteregian Hills predate ~114 Ma. The best "fit" for the pattern would be at ~120 Ma, where the paleogeomagnetic field was reversing polarity about every 500 000 years. Thus, intrusion of the Monteregian Hills likely happened over a relatively short time interval of ~ 2 Ma and certainly of < ~5 Ma.

The duration of plutonic emplacement places some limitations on the mode of origin of the Monteregian intrusions. One possibility is that they represent a "failed hot spot", i.e., one that led to the emplacement of alkalic plutons at depth but insufficiently intense to break through to extrude the diagnostic alkaline lavas. The rifting tectonics found in the Monteregians fits with the hot spot hypothesis (Wilson 1973). Using data from LePichon and Fox (1971), the North American plate at the Monteregians was moving westward at ~1.4 cm/year, assuming a stationary Mid-Atlantic Ridge. The westward motion fits with the Monteregian petrologic trends. The Monteregians span ~120 km excluding Mt. Megantic because of the paleomagnetic data gap between it and the other intrusives. Using an intrusive period of 2 Ma, this gives a rate of motion of ~6 cm/year, assuming the hot spot plume was stationary. The discordance in the rates leads to a number of speculations. Firstly, the polarity pattern may not be sufficiently well defined. If so, then using the 5 Ma limit still gives a too large ~2.4 cm/year rate of motion. Also, the consistency and density of site polarities along the belt does not fit well with this notion. Secondly, the plume itself may have been moving generally eastward at ~4.5 cm/year. Morgan (1972) provides some evidence that plumes are not always fixed points, especially those under continents. The ~25° angular difference between the spreading direction (e.g., the Kelvin Fracture Zone) and Monteregian trend would support this notion. It would also support the third speculation that the plutonic source migrated eastward in response to, rather than because of, progressive opening of a tensional rift belt. If true, then the Monteregian intrusives reflect minor adjustments in the middle of the North American plate corresponding to the onset of the Laramide Orogeny. The 120 Ma hairpin in the apparent polar wander curve for North America (Irving 1979) signals the probable presence of such adjustments.

Conclusions

(1) The Oka, Carillon (excepting one site), Ile Cadieux, Brilund, and Ste. Dorothée intrusions at the west end of the Monteregian Hills retain a stable normal primary magnetization acquired during cooling in one polarity interval ~120 Ma ago.

(2) Using AF step demagnetization of every specimen to determine those with reliable "end point" directions so that these can be combined to get site mean directions did not improve the site mean dispersion statistics beyond the dispersion statistics obtained from the conventional "pilot" specimen method.

(3) Normal and reversed sites from the Monteregian Hills give statistically identical antiparallel mean remanence directions. The resulting pole position based on all sites is ~1.5° from that reported by Larochelle (1969) for his predominantly reversed suite of sites. This pole at 169.0°W, 72.4°N ($\delta_p = 2.8°$, $\delta_m = 3.7°$) is consistent with an age of 120 ± 4 Ma for the Monteregian Hills.

(4) The polarity pattern along the Monteregian Hills belt suggests that they were emplaced during four polarity intervals lasting ~2 Ma about 120 Ma ago. Emplacement began with explosive ultrabasic–basic intrusion at the west end of the belt during a normal geomagnetic interval. The intrusive centre progressed eastwards through a reversed, a normal, and a final reversed polarity interval at the east end by which time the plutons were dominantly acidic and being passively emplaced. The paleomagnetic data is most consistent with the hypotheses that the Monteregian intrusives record the westerly motion of the North American plate over an easterly moving hot spot, or the progressive easterly rifting of the plate in response to regional stresses during the Laramide Orogeny.

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